

# The Factorial-Denominator Series: Complete Reasoning Record

*Erdős Problem #68*

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## ABSTRACT

Write  $d_n = n! - 1$ ,  $H_M = \sum_{2 \leq n \leq M} d_n^{-1}$  and  $L_M = \text{lcm}(d_2, \dots, d_M)$ . For a prime  $p$  with  $e = v_p(L_M) \geq 1$ , let  $J = \{n : 2 \leq n \leq M, v_p(d_n) = e\}$  and write  $d_n = p^e u_n$  on  $J$ . Then

$$v_p(\text{den}(H_M)) = e \iff \sum_{n \in J} u_n^{-1} \not\equiv 0 \pmod{p},$$

the inverses being taken modulo  $p$ . The prefixes through 138 and 2592 cancel the primes 139 and 2593 completely. Second, the common denominator itself grows fast:

$$\liminf_{N \rightarrow \infty} \frac{\log L_N}{N^{3/2} \log N} \geq \frac{2\sqrt{2}}{3},$$

by an elementary segment argument that uses  $\gcd(i! - 1, j! - 1) \mid j!/i! - 1$  and no external multiplicity theorem. The two statements sit either side of reduction, and the cancellations at 139 and 2593 are why the second cannot be substituted for the first.

Wilson's theorem gives arbitrarily late first occurrences of primes among the factorial gaps, so prefix-private prime support of the denominators occurs cofinally, with no bound relating the prime to its first hit. Four exact criteria locate the irrationality of  $S = \sum_{n \geq 2} d_n^{-1}$ : a strict-successor carry equivalence, a companion-orbit residue condition, a lower-interval escape condition of width  $O(m^{-2})$ , and a boundary covering the whole shifted family  $\sum_{n \geq 2} 1/(n! + t)$  for every integer  $t \geq -1$ , whose member  $t = 0$  returns the irrationality of  $e$ . A finite-channel radius bound gives  $3t^3 < 2(R + 1)$  under the exact cancellation and factorial-size hypotheses, and the growth statement above raises the asymptotic constant in that estimate from  $3/2$  to  $16/9$ . Two finite computations exclude denominators:  $q \nmid 299999!$  from an exact interval carry census through 300000, and the kernel-checked bounds  $q \geq 2^{39990}$  and  $q > 10^{12040}$  from a certified continued-fraction enclosure. The factorial-grid exclusion uses the external GMP computation and remains a separate outstanding Lean obligation. The irrationality of  $S$  is open, and the remaining arithmetic inputs are stated in the last section.

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\* *Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

# 1 Which prime powers survive reduction

**Problem 1.1** (Erdős #68). Is  $S = \sum_{n \geq 2} (n! - 1)^{-1}$  irrational?

Erdős states the question on p. 102 of his 1988 survey and, in the same passage, records the expectation that  $\sum_n 1/(n! + t)$  is irrational, indeed transcendental, for every integer  $t$  [1, p. 102]. That is conjectural context. Numbering and current status follow [Bloom's Erdős problem catalogue](#) [2]; the problem is open.

Put

$$d_n = n! - 1, \quad H_M = \sum_{n=2}^M \frac{1}{d_n}, \quad L_M = \text{lcm}(d_2, \dots, d_M), \quad A_M = \sum_{n=2}^M \frac{L_M}{d_n},$$

so that  $H_M = A_M/L_M$ . Throughout,  $\text{den}(x)$  is the positive denominator of a rational number in lowest terms. The first question is which prime power present in  $L_M$  remains in  $\text{den}(H_M)$ .

**Theorem 1.2** (maximal prime-power survival). *Let  $M \geq 2$ , let  $p$  be a prime dividing  $L_M$ , and put  $e = v_p(L_M)$ . Let  $J = \{n : 2 \leq n \leq M, v_p(d_n) = e\}$  and write  $d_n = p^e u_n$  for  $n \in J$ . Then, with inverses in  $\mathbb{F}_p$ ,*

$$v_p(\text{den}(H_M)) = e \iff \sum_{n \in J} u_n^{-1} \neq 0 \text{ in } \mathbb{F}_p. \quad (1)$$

*Proof.* Write  $L_M = p^e W$  with  $p \nmid W$ . If  $v_p(d_n) < e$ , then  $p \mid L_M/d_n$ . If  $n \in J$ , then  $u_n(L_M/d_n) = W$ , so  $L_M/d_n \equiv W u_n^{-1} \pmod{p}$ . Summing over  $2 \leq n \leq M$  gives

$$A_M \equiv \frac{L_M}{p^e} \sum_{n \in J} u_n^{-1} \pmod{p}. \quad (2)$$

The historical manuscript cites Erdos68/PaperCompletePrimePole.lean, line 118. This module is absent from the retained source pin; this citation does not establish kernel verification at that pin.

Finally  $\text{den}(H_M) = L_M / \gcd(A_M, L_M)$ , whose  $p$ -valuation is  $e$  exactly when  $p \nmid A_M$ . Since  $W$  is invertible modulo  $p$ , that is exactly the stated condition.  $\square$

The criterion decides the survival of the full exponent  $e$ . A zero residue gives  $v_p(\text{den}(H_M)) < e$ , and further congruences are needed to determine the surviving exponent. The proof uses no property of factorials beyond the displayed denominators, so it applies to any finite sum of reciprocals. It is the first valuation layer of the reciprocal-sum formula of Louwsma and Martino [3, Lemma 4.1, p. 10]. What is specific here is the behaviour of the actual family  $d_n = n! - 1$ , and that behaviour is not generic.

**Two complete cancellations.** Take  $M = p - 1$ . The indices below with  $p \mid d_n$  are all of them, and every one has valuation exactly one.

| $p$  | indices $n$     | $d_n/p \pmod{p}$ | inverses modulo $p$ |
|------|-----------------|------------------|---------------------|
| 139  | 69, 122, 137    | 6, 49, 73        | 116, 122, 40        |
| 2593 | 349, 2243, 2591 | 1508, 1566, 1678 | 1367, 356, 870      |

The inverse sums are  $278 = 2 \cdot 139$  and  $2593$ . By Theorem 1.2,  $139 \nmid \text{den}(H_{138})$  and  $2593 \nmid \text{den}(H_{2592})$ : each prime divides the common denominator of its prefix and vanishes from the reduced one. Both hit lists and both valuations are checked by the recurrence

$$r_1 = 1, \quad r_n \equiv nr_{n-1} \pmod{p^2}, \quad 0 \leq r_n < p^2,$$

run through  $n = p - 1$ . A hit is an index with  $r_n \equiv 1 \pmod{p}$ , its lifted cofactor is  $(r_n - 1)/p$  modulo  $p$ , and valuation two would mean  $r_n = 1$ , which occurs at neither prime. That enumeration is a finite calculation separate from the theorem.

A prime is *prefix-private at  $m$*  when it divides  $d_m$  and divides none of  $d_2, \dots, d_{m-1}$ .

**Proposition 1.3** (cofinal first prime occurrences). *For every integer  $B \geq 0$  there are a prime  $q$  and an integer  $m > B$  with  $m < q$ ,  $q \mid m! - 1$  and  $\gcd(q, k! - 1) = 1$  for every  $k$  with  $2 \leq k < m$ .*

*Proof.* Choose a prime  $q \geq B! + 5$ . Wilson's theorem gives  $(q - 2)! \equiv 1 \pmod{q}$ , so the set of indices  $n \geq 2$  with  $q \mid n! - 1$  is nonempty and contains no element above  $q - 2$ ; let  $m$  be its least element. If  $m \leq B$ , then  $q \leq m! - 1 \leq B! - 1$ , contradicting the choice of  $q$ . Hence  $m > B$ , and minimality gives the coprimality assertions.  $\square$

The construction gives no useful upper bound for  $q$  in terms of its first hit, and that comparison is what the scale inequalities of §4 consume. Wilson reflection limits what can be inferred from a large prime factor alone: for odd  $n < q$  with  $q \mid n! - 1$ , the identity  $(q - 1 - n)! n! \equiv (-1)^{n+1} \pmod{q}$  gives a second hit at  $q - n - 1$ . The reflection identity is classical; see Stewart [7, p. 462, (4)].

*Status.* The problem treated here is open, and this note does not close it. Every statement below marked as checked is a proposition that the pinned Lean kernel accepts from the sources this note links to, with no sorry, no added axiom, and no unchecked evaluation. That is a claim about the formal statement, not about its mathematical interest, its novelty, or the original problem. The unresolved obligations are named exactly, in their own section, and none of the finite computations, reductions, or no-go results here removes one of them.

*Companion system context.* The [claim and trust boundary](#), [cold-clone route to proof authority](#), and [public contribution protocol](#) are described in sibling papers. Those descriptions do not change the mathematical status of this note.

## 2 The growth of the common denominator

Theorem 1.2 concerns the denominator after reduction. The common denominator before reduction admits an unconditional lower bound of its own, by an elementary argument.

**Lemma 2.1** (product, least common multiple, pairwise gcd). *For positive integers  $x_1, \dots, x_k$ ,*

$$\prod_{i=1}^k x_i \mid \text{lcm}(x_1, \dots, x_k) \prod_{i < j} \gcd(x_i, x_j).$$

*Proof.* Fix a prime  $r$  and relabel so that  $a_1 \leq \dots \leq a_k$ , where  $a_i = v_r(x_i)$ . The right-hand side has  $r$ -valuation  $a_k + \sum_{i < j} \min(a_i, a_j) = a_k + \sum_{i=1}^{k-1} (k-i)a_i$ , and the left-hand side has  $\sum_{i=1}^k a_i$ . The difference is  $\sum_{i=1}^{k-1} (k-i-1)a_i$ , which is nonnegative.  $\square$

**Lemma 2.2** (factorial-gap gcd). *For  $2 \leq i < j$ , the integer  $g = \gcd(i! - 1, j! - 1)$  divides  $j!/i! - 1$ , and  $g \leq j!/i! - 1 < j^{j-i}$ .*

*Proof.* Both  $i!$  and  $j!$  are congruent to 1 modulo  $g$ . The quotient  $j!/i! = (i+1)(i+2)\dots j$  is an integer, and  $j! = i! \cdot (j!/i!)$ , so  $j!/i! \equiv 1 \pmod{g}$ . Since  $j!/i! \geq i+1 > 1$ , the positive integer  $j!/i! - 1$  is a multiple of  $g$ , whence  $g \leq j!/i! - 1$ . Finally  $j!/i!$  is a product of  $j-i$  integers each at most  $j$ .  $\square$

**Lemma 2.3** (segment inequality). *For  $2 \leq k \leq N-1$ ,*

$$\sum_{n=N-k+1}^N \log(n! - 1) \leq \log L_N + \binom{k+1}{3} \log N. \quad (3)$$

*The historical manuscript cites Erdos68/PaperCompleteGcdSegment.lean, line 28. This module is absent from the retained source pin; this citation does not establish kernel verification at that pin.*

*Proof.* Apply Lemma 2.1 to  $x_n = d_n$  for  $N-k+1 \leq n \leq N$ . Their least common multiple divides  $L_N$ . By Lemma 2.2 each pairwise gcd is smaller than  $N^{j-i}$ , and over a block of  $k$  consecutive indices

$$\sum_{i < j} (j-i) = \sum_{d=1}^{k-1} d(k-d) = \binom{k+1}{3}.$$

Taking logarithms of the resulting divisibility gives (3).  $\square$

**Theorem 2.4** (common-denominator growth).

$$\liminf_{N \rightarrow \infty} \frac{\log L_N}{N^{3/2} \log N} \geq \frac{2\sqrt{2}}{3}.$$

*Proof.* Fix  $\alpha > 0$  and take  $k = \lfloor \alpha\sqrt{N} \rfloor$ , which satisfies  $2 \leq k \leq N-1$  for all large  $N$ . Put  $u = N - k + 1$ .

For the left-hand side of (3), use  $n! - 1 \geq n!/2$  and  $\log n! \geq n \log n - n$ . Since  $n \mapsto n \log n - n$  increases,

$$\sum_{n=u}^N \log(n! - 1) \geq k(u \log u - u) - k \log 2.$$

Here  $u \geq N - \alpha\sqrt{N}$  and  $k = \alpha\sqrt{N} + O(1)$ , so  $ku \log u = \alpha N^{3/2} \log N + O(N \log N)$ , while  $ku = O(N^{3/2})$  and  $k \log 2 = O(\sqrt{N})$ . Hence the left-hand side is  $\alpha N^{3/2} \log N + o(N^{3/2} \log N)$ .

For the right-hand side,  $\binom{k+1}{3} \leq (k+1)^3/6$ , so  $\binom{k+1}{3} \log N = \frac{\alpha^3}{6} N^{3/2} \log N + o(N^{3/2} \log N)$ . Therefore

$$\liminf_{N \rightarrow \infty} \frac{\log L_N}{N^{3/2} \log N} \geq \alpha - \frac{\alpha^3}{6}.$$

The right-hand side is maximised at  $\alpha = \sqrt{2}$ , with value  $\sqrt{2} - \frac{2\sqrt{2}}{6} = \frac{2\sqrt{2}}{3}$ .  $\square$

The LCM asymptotic is proved here by an ordinary mathematical argument and is not kernel-checked. The associated asymptotic Lean files are proof candidates and are not being cited as kernel verification. The finite-block inequality has a separate Lean source, [finite terminal-block inequality](#). Its two ingredients are Lemmas 2.1 and 2.2, and it needs no factorial-congruence multiplicity theorem. A weaker exponent 4/3 follows instead from Theorem 12 of Garaev, Luca and Shparlinski [6, arXiv v1, Thm. 12, p. 16], which bounds by  $O(N^{2/3})$  the number of solutions of  $n! \equiv a \pmod{r}$  with  $r$  prime,  $a \not\equiv 0 \pmod{r}$ , and  $H+1 \leq n \leq H+N$  where  $0 \leq H < H+N < r$ ; that deduction is recorded in the complete reasoning record and is superseded here. No antecedent for a lower bound on the least common multiple of the numbers  $n! - 1$  was located, and the searches were for factorial values shifted by one; least common multiples of consecutive integers and of linear recurrences are a different family. The finding is that the statement is possibly known and was not located.

The constant  $2\sqrt{2}/3$  is the best this argument gives. The gain from a block of  $k$  indices is largest at the top of the range and the pairwise-gcd loss grows with the spread, so consecutive top indices are simultaneously optimal for both terms, and the bound  $j!/i! < N^{j-i}$  is sharp to first order there. Convergence is slow: the omitted term  $-ku$  is of order  $N^{3/2}$ , so the finite ratio falls short of the limit by about  $\sqrt{2}/\log N$ .

Theorem 2.4 is a statement about  $L_N$  and not about  $\text{den}(H_N)$ . Consecutive denominators  $(N-1)! - 1$  and  $N! - 1$  are already coprime for  $N \geq 3$ , so the qualitative failure of common-denominator clearing follows from two terms; the estimate is the quantitative growth statement. Theorem 1.2 and the cancellations at 139 and 2593 show why the substitution fails: a prime may occupy the common denominator and be absent from the reduced one. One cancelled prime is not an asymptotic cancellation theorem. Bounding the reduced denominator from below at every index is a stronger assertion, and it is exactly what an irrationality proof through the clearing-scale route needs. That stronger assertion has not been shown to follow from rationality of  $S$  alone; it would require an extra small-tail hypothesis such as  $d_N(S - H_N) < 1$ .

### 3 Carries and the rationality boundary

Define the strict integer successor and the predecessor gap by

$$Z_m = \lfloor m!H_m \rfloor + 1, \quad \Delta_m = Z_{m-1} - (m-1)!H_{m-1} \quad (m \geq 3),$$

so  $0 < \Delta_m \leq 1$ , and define the integer carry  $b_m$  by

$$Z_m = mZ_{m-1} + 1 - b_m. \tag{4}$$

The historical manuscript cites Erdos68/PaperCompleteGcdSegment.lean, line 73. This module is absent from the retained source pin; this citation does not establish kernel verification at that pin.

Call  $b_m = 1$  a *unit carry*. Put  $E_m = m!(S - H_m)$  and  $\varepsilon_m = 1/(m! - 1)$ . Since  $d_{n+1} > (n+1)d_n$ , a geometric majorant gives the tail estimate

$$0 < E_m < \frac{2m!}{(m+1)! - 1} < \frac{2}{m} \leq 1 \quad (m \geq 2). \quad (5)$$

**Theorem 3.1** (strict-successor characterisation). *For  $m \geq 3$ ,*

$$b_m = 1 \iff m \mid Z_m \iff 1 + \varepsilon_m < m\Delta_m \leq 2 + \varepsilon_m. \quad (6)$$

The historical manuscript cites Erdos68/PaperCompleteExisting.lean, line 79. This module is absent from the retained source pin; this citation does not establish kernel verification at that pin.

Moreover

$$S \in \mathbb{Q} \iff b_m = 1 \text{ for all sufficiently large } m, \quad (7)$$

$$S \notin \mathbb{Q} \iff \forall B \exists m > B : m \nmid Z_m. \quad (8)$$

If  $S = a/q$  with  $a \in \mathbb{Z}$ ,  $q \geq 1$  and  $b_m \neq 1$ , then  $q \nmid (m-1)!$  and  $q \geq m$ .

*Proof.* From  $m!H_m = mZ_{m-1} - m\Delta_m + 1 + \varepsilon_m$  one gets  $b_m = \lceil m\Delta_m - 1 - \varepsilon_m \rceil$  with  $-1 \leq b_m \leq m-1$ , which gives both equivalences in (6), endpoints included.

Suppose  $S = a/q$  and  $q \mid (m-1)!$ . For  $j = m-1$  and  $j = m$  the number  $j!S$  is an integer, and  $j!H_j = j!S - E_j$  with  $0 < E_j < 1$  by (5), so  $Z_j = j!S$ . Substituting into (4) forces  $b_m = 1$ . Every fixed  $q$  divides  $(m-1)!$  eventually, which gives one direction of (7) and, by contraposition, the final assertion;  $q < m$  would imply  $q \mid (m-1)!$ . Conversely, an eventual unit-carry tail makes  $Z_m/m!$  eventually constant, and  $0 < Z_m/m! - H_m \leq 1/m!$  with  $H_m \rightarrow S$  identifies that constant as  $S$ , which is then rational. Negating (7) gives (8).  $\square$

The divisibility conclusion retains prime-power multiplicities, so it constrains valuations, prime support included. It does not require a prime divisor of  $q$  larger than  $m-1$ .

### 3.1 Why one proposed window argument is circular

One route to cofinal non-unit carries tries to exclude two consecutive unit carries by extracting a prime-power or quotient-gcd obstruction from their cleared width-one window. The exact two-step recurrence shows what that window can deliver. Let  $U_k$  and  $V_k$  be the numerator and denominator of the reduced predecessor gap and let  $G_k > 0$  be the exact transition normaliser. For  $m \geq 3$ , the two unit carries are equivalent to an integer offset  $\Omega_m$  with  $0 < \Omega_m \leq D_m$ , the window denominator telescopes independently of the carry values,

$$D_m = V_{m+2}G_{m+1}G_m = V_m(m! - 1)((m+1)! - 1),$$

and under the pair assumption the offset factors to match,  $\Omega_m = U_{m+2}G_{m+1}G_m$ . Cancelling the common positive normaliser reduces the complete window to

$$0 < U_{m+2} \leq V_{m+2},$$

which every reduced positive gap in  $(0, 1]$  already satisfies. For that specified window, therefore, an obstruction obtained only after imposing the carry pair rewrites a bound that holds anyway. The calculation does not disprove adjacent unit carries and does not exclude an independent restriction on the transition numerators or their valuations, which would need its own argument constraining the predecessor state before the pair is assumed.

### 3.2 The companion coordinate

Let  $C = \sum_{n \geq 2} (n!(n! - 1))^{-1}$ . The identity  $1/(n! - 1) = 1/n! + 1/(n!(n! - 1))$  and absolute convergence give

$$S = C + e - 2. \quad (9)$$

The canonical factorial digits of a real  $x$  are  $a_m(x) = \lfloor m!x \rfloor - m \lfloor (m-1)!x \rfloor \in \{0, \dots, m-1\}$ , and  $x$  is rational exactly when they vanish from some index on. That criterion is due to Cantor [4] and is recorded in Galambos [5, Ch. 1].

**Proposition 3.2** (companion-orbit criterion).

$$S \in \mathbb{Q} \iff \lfloor m!C \rfloor \equiv -2 \pmod{m} \text{ for all sufficiently large } m.$$

*Proof.* Put  $J_m = \sum_{k=0}^m m!/k!$ . Then  $J_m$  is an integer, every summand with  $k < m$  is divisible by  $m$  and the summand at  $k = m$  is 1, so  $J_m \equiv 1 \pmod{m}$ ; and  $0 < m!e - J_m < 1$  for  $m \geq 2$ .

Suppose  $S$  is rational. For all large  $m$  both  $(m-1)!S$  and  $m!S$  are integers, so  $m \mid m!S$ . Multiplying (9) by  $m!$  and using  $\lfloor -m!e \rfloor = -J_m - 1$  gives  $\lfloor m!C \rfloor = m!S + 2m! - J_m - 1 \equiv -2 \pmod{m}$ .

Conversely, assume the congruence from some index on. Because  $0 \leq a_m(C) < m$ , for  $m \geq 3$  the congruence is equivalent to  $a_m(C) = m - 2$ . Choose  $N$  beyond the exceptional indices. The canonical expansion gives

$$C = \frac{\lfloor N!C \rfloor}{N!} + \sum_{m > N} \frac{m-2}{m!},$$

and the telescope  $\sum_{m > N} (m-1)/m! = 1/N!$  turns this into

$$S = C + e - 2 = \frac{\lfloor N!C \rfloor + 1}{N!} + \sum_{m=2}^N \frac{1}{m!},$$

which is rational. □

### 3.3 Escape from a smaller interval

Write  $\theta_m = \{m!S\}$ . The exact lower interval has width  $E_m/m \in (0, 2/m^2)$ , so the target shrinks quadratically.

**Proposition 3.3** (lower-interval criterion).

$$S \notin \mathbb{Q} \iff \forall B \exists m > B : E_m \leq m\theta_{m-1}. \quad (10)$$

For  $m \geq 3$  the finite condition

$$m\Delta_m \leq 1 + \varepsilon_m \quad \text{or} \quad 1 + \varepsilon_m + \frac{2}{m} \leq m\Delta_m \quad (11)$$

implies the escape inequality in (10). Cofinally many instances of (11) therefore imply  $S \notin \mathbb{Q}$ .

*Proof.* If  $S$  is rational, the remainders  $\theta_{m-1}$  vanish eventually while  $E_m > 0$ , so the escape inequality fails eventually. Conversely, suppose  $m\theta_{m-1} < E_m$  for every sufficiently large  $m$ . By (5) this gives  $m\theta_{m-1} < 1$ , so  $a_m(S) = 0$  and  $\theta_m = m\theta_{m-1}$ . A positive remainder grows past one under that recurrence, so the remainders vanish and  $S$  is rational.

For the finite implication, suppose  $m\theta_{m-1} < E_m$ . The tail recurrence is  $mE_{m-1} = 1 + \varepsilon_m + E_m$ , so  $E_{m-1} > E_m/m > \theta_{m-1} \geq 0$ , and with  $0 < E_{m-1} < 1$  the strict-successor identity gives  $\Delta_m = E_{m-1} - \theta_{m-1}$ . Hence

$$1 + \varepsilon_m < m\Delta_m = 1 + \varepsilon_m + E_m - m\theta_{m-1} \leq 1 + \varepsilon_m + E_m < 1 + \varepsilon_m + \frac{2}{m},$$

which contradicts (11).  $\square$

Escape at a single index does not give a non-unit carry there. The exact classification is

$$E_m \leq m\theta_{m-1} \iff b_m \neq 1 \text{ or } a_m(S) = m - 1 \quad (m \geq 3), \quad (12)$$

The historical manuscript cites Erdos68/PaperCompleteExisting.lean, line 104. This module is absent from the retained source pin; this citation does not establish kernel verification at that pin.

so the maximal-digit branch is available. It is realised: at  $m = 52$  exact rational arithmetic gives  $b_{52} = 1$  together with  $52\Delta_{52} > 1 + \varepsilon_{52} + 2/52$ , the margin being about 0.5689674908. Since 52 is composite, this settles nothing about prime indices, and the cofinal statement in Proposition 3.3 does not identify escape with a non-unit carry.

### 3.4 The whole shifted family

Erdős asked about  $S$  inside a family, and the same coordinate covers the family. For an integer  $t \geq -1$  put  $S_t = \sum_{n \geq 2} 1/(n! + t)$  and  $C_t = \sum_{n \geq 2} 1/(n!(n! + t))$ , so that  $S_t = -tC_t + (e - 2)$ .

**Theorem 3.4** (uniform family boundary). *For every integer  $t \geq -1$ , the series  $S_t$  is rational exactly when*

$$[tm!C_t] \equiv 2 \pmod{m}$$

*for all sufficiently large  $m$ , and irrational exactly when that residue is missed cofinally. The member  $t = -1$  is  $S$ , and the member  $t = 0$  is  $e - 2$ , whose orbit is constantly 0 and therefore misses the residue class at every  $m \geq 3$ , so the boundary returns the irrationality of  $e$ .*



The  $t = 0$  specialisation recovers a classical fact and is included because it is the one member of the family the criterion decides, which fixes the sense in which the criterion is exact. For every other  $t$  the producer of cofinal misses is missing, and the family is open.

## 4 From denominator structure to a scale criterion

For an integer  $p \geq 3$  put  $I_p = \{2, \dots, 2p-1\}$  and  $F_p = (p-1)!$ , and define

$$\begin{aligned} D_p &= \text{lcm}_{\substack{i,j \in I_p \\ i < j}} \gcd(d_i, d_j), & C_p &= \text{lcm}(F_p, D_p), \\ L_p^{\text{blk}} &= \text{lcm}(F_p, d_2, \dots, d_{2p-1}), & R_p &= L_p^{\text{blk}} / C_p. \end{aligned}$$

The normalised core is  $\tilde{C}_p = C_p / F_p$ , so that

$$L_p^{\text{blk}} = C_p R_p = F_p \tilde{C}_p R_p. \quad (13)$$

The historical manuscript cites Erdos68/PaperCompleteExisting.lean, line 125. This module is absent from the retained source pin; this citation does not establish kernel verification at that pin.

The factorial base  $F_p$  belongs inside every one of these quantities, and omitting it changes the scale comparisons below by a factorial factor. Put

$$T_p = \sum_{i \in I_p} \frac{L_p^{\text{blk}}}{d_i}, \quad \rho_p = (-T_p) \bmod R_p, \quad K_p = 2p^2(2p-1)!,$$

with  $\rho_p$  the least nonnegative representative. All of these depend only on a finite prefix.

The core records every prime-power layer occurring in at least two gaps, together with the base. A prime dividing  $R_p$  therefore has a unique maximal-valuation gap above the base valuation, so the argument of Theorem 1.2 applies to it and gives  $\gcd(T_p, R_p) = 1$ . This is where the first theorem returns. Consequently, when  $R_p > 1$ , the ratio  $T_p / R_p = C_p H_{2p-1}$  is not an integer and

$$\frac{\rho_p}{R_p} = \lceil C_p H_{2p-1} \rceil - C_p H_{2p-1} \in (0, 1), \quad (14)$$

a gap to the next integer.

**Theorem 4.1** (complementary-residue criterion). *Suppose that for every  $B$  there is a prime  $p > B$  with  $R_p > 1$  and*

$$(2p+1)L_p^{\text{blk}} < K_p \rho_p. \quad (15)$$

*Then  $S$  is irrational.*

*Proof.* Suppose  $S = a/q$  with  $q \geq 1$ , and take a block with  $p > \max(q, 3)$  satisfying the hypotheses. Then  $q \mid F_p$ , so  $U = L_p^{\text{blk}}(S - H_{2p-1})$  is a positive integer, and  $L_p^{\text{blk}}S =$

$R_p(C_p S)$  is a multiple of  $R_p$ ; hence  $U \equiv -T_p \pmod{R_p}$  and  $U \geq \rho_p$ . For the other side,  $1/(n! - 1) < 2/n!$  for  $n \geq 3$ , and a geometric majorant gives

$$0 < S - H_{2p-1} < \frac{2}{(2p)!} \cdot \frac{2p+1}{2p} = \frac{2p+1}{K_p}.$$

Therefore  $K_p \rho_p \leq K_p U < (2p+1)L_p^{\text{blk}}$ , contrary to (15).  $\square$

Cancelling  $R_p$  in (15) through (13) displays the quantitative issue directly:

$$(2p+1)C_p < K_p \frac{\rho_p}{R_p}. \quad (16)$$

The private modulus has gone. A supply of large private factors therefore supplies neither a small collision core nor a lower bound for the normalised gap  $\rho_p/R_p$ , which is at most one. That is the exact reason Proposition 1.3 does not close the argument.

The moving-factor criterion is a more specialised sufficient condition. For a prefix-private prime  $q$  at  $m \geq 4$  the tailored block parameter is  $p = \lfloor m/2 \rfloor + 1$ , which need not itself be prime. For the factors 1 and  $q$  of the private modulus the factor-pair floor is  $\min(\rho_p, R_p/q)$ , and its scale inequality splits exactly:

$$(2p+1)L_p^{\text{blk}} < K_p \min(\rho_p, R_p/q) \iff \begin{cases} (2p+1)L_p^{\text{blk}} < K_p \rho_p, \\ (2p+1)C_p q < K_p. \end{cases} \quad (17)$$

The historical manuscript cites Erdos68/PaperCompleteSupplementary.lean, line 59. This module is absent from the retained source pin; this citation does not establish kernel verification at that pin.

Testing both entries of the minimum and using (13) gives the equivalence, so the reduction has no opaque floor premise left. Cofinal instances of both inequalities imply irrationality. The single global condition of Theorem 4.1 is separately sufficient, so the pair is one sufficient route among several.

One structural limitation is worth recording beside these criteria. If the owner index  $n$  is fixed and  $p > n! - 1$ , then  $n! - 1$  divides  $(p-1)!$ , so its private quotient in the block at  $p$  is exactly one. Large private quotients seen at small blocks, such as the factor 719 owned at  $n = 6$ , are therefore finite-range phenomena, and the selected factors must escape with  $p$ .

## 5 A limitation of finite-channel cancellation

For a finitely supported integer vector  $c = (c_i)$  on indices  $i \geq 2$ , put

$$M(c) = \sum_i c_i i!, \quad W_{d,i} = \frac{i!}{(d!)^{\lfloor i/d \rfloor}}, \quad V_d(c) = \sum_i c_i W_{d,i} \quad (d \geq 2).$$

Each  $W_{d,i}$  is an integer: writing  $i = kd + r$  with  $0 \leq r < d$ , the quotient  $i!/((d!)^k r!)$  is a multinomial coefficient. Since  $i! = (d!)^{\lfloor i/d \rfloor} W_{d,i}$  and  $d! \equiv 1 \pmod{d! - 1}$ ,

$$V_d(c) \equiv M(c) \pmod{d! - 1}. \quad (18)$$

**Theorem 5.1** (integral normal form). *For every finite integer support and every  $d \geq 2$  there is an integer  $k$  with  $V_d(c) = M(c) + (d! - 1)k$ .*

**Theorem 5.2** (quotient-band breakpoint). *Let  $d \geq 2$  and  $k \geq 0$ , and suppose every supported index  $i$  satisfies  $kd \leq i < (k+1)d$ . Then  $M(c) = (d!)^k V_d(c)$ . In particular, cancellation in the first band  $d \leq i < 2d$  forces  $M(c) = 0$ ; and if every supported index is at least  $d$  while  $M(c) \neq 0$  and  $V_d(c) = 0$ , then some supported index is at least  $2d$ .*

*Proof.* Within one band the quotient  $[i/d]$  is the constant  $k$ , so  $i! = (d!)^k W_{d,i}$  for every supported index and the factor  $(d!)^k$  comes out of the sum. The two consequences follow by taking  $k = 1$  and by contraposition.  $\square$

The hard step is the constant quotient; no valuation estimate enters. A nonzero moment therefore cannot be hidden entirely below  $2d$  while the  $d$ -channel cancels. This is a finite-family obstruction: it constructs no cancelling family and says nothing about simultaneous channels or residual size.

Consequently a vanishing  $d$ -th channel forces  $(d! - 1) \mid M(c)$ , and annihilating every channel  $2 \leq d \leq D$  forces  $L_D \mid M(c)$ , where  $L_D = \text{lcm}_{2 \leq d \leq D} (d! - 1)$  is the same quantity as in §2. Every zero-moment variation of the support changes a normalised channel contribution by an integer only, so such variations cannot manufacture an extra fractional cancellation coordinate.

### 5.1 The exact low-channel lattice and residual class

The integral divisor basis makes the preceding divisibility statement exact. Let  $U_n$  be the isolated  $n$ -channel unit, let  $K_D$  be the canonical vector with moment  $L_D$  and channels  $2, \dots, D$  equal to zero, and write  $a_D = [e_1]K_D$  and  $u_n = [e_1]U_n$ . Every vector with zero initial coordinate and vanishing channels through  $D$  has a unique expression

$$c = tK_D + \sum_{n>D} z_n U_n, \quad M(c) = tL_D, \quad (19)$$

with finitely many nonzero integers  $z_n$ . Requiring support on  $n \geq 2$  adds precisely the scalar equation

$$ta_D + \sum_{n>D} z_n u_n = 0. \quad (20)$$

**Theorem 5.3** (exact attainable moment ideal). *Fix  $D \geq 2$  and a prime  $p$  with  $D/2 < p \leq D$ . Put*

$$H = D(2p - 1), \quad G_D = \gcd\{|u_n| : D < n \leq H\}, \quad \mu_D = L_D \frac{G_D}{\gcd(G_D, a_D)}.$$

*The moments of finite integer vectors supported on  $n \geq 2$  and cancelling all channels  $2, \dots, D$  are exactly  $\mu_D \mathbb{Z}$ . The integer  $\mu_D$  is positive, is independent of the eligible prime  $p$ , and is attained by a vector of coefficient content one, hence by a primitive vector.*

*Proof.* The finite-horizon theorem gives  $G_D > 0$  and identifies it with the gcd of the whole tail  $\{u_n : n > D\}$ ; moreover  $H < 2D^2$ . Thus the finite integer combinations in (20) form  $G_D \mathbb{Z}$ , so that the equation is soluble exactly when  $G_D \mid ta_D$ . Dividing by

$\gcd(G_D, a_D)$  shows that  $t$  is a multiple of  $G_D / \gcd(G_D, a_D)$ , and (19) gives the displayed moment ideal. Bezout coefficients attain its positive generator. If an attaining vector had a nontrivial common coefficient divisor, division by that divisor would produce a smaller positive attainable moment. Hence its content is one. The ideal itself is intrinsic, so two eligible primes give the same positive generator.  $\square$

For

$$\mathcal{R}(c) = \sum_{d \geq 2} \frac{V_d(c)}{d! - 1}, \quad H_D = \sum_{2 \leq d \leq D} \frac{1}{d! - 1},$$

the same coordinates retain the real information that the moment ideal alone does not see.

**Theorem 5.4** (full residual transparency). *For the vector in (19),*

$$\mathcal{R}\left(tK_D + \sum_{n > D} z_n U_n\right) = tL_D(S - H_D) + \sum_{n > D} z_n.$$

*The residual series converges for every finite vector supported away from index zero. A zero-moment vector has integral residual, and two such vectors with the same factorial moment have residuals differing by an integer.*

*Proof.* For  $d > D$ , the  $K_D$  channel is  $L_D$ , while each  $U_n$  contributes one to its isolated residual coordinate. Termwise decomposition therefore gives the factorial-gap tail  $L_D(S - H_D)$  plus the finite coordinate mass  $\sum z_n$ . The factorial-gap tail is summable and the coordinate correction has finite support, which proves convergence. In the full divisor-basis coordinates the coefficient of the  $e_1$  column equals the factorial moment; when that moment is zero only the integral coordinate mass remains. Subtracting two equal-moment identities proves the last assertion.  $\square$

The exact Lean endpoint packages are [primitive attainment of the moment-ideal generator](#) and [the equal-moment residual class](#). They determine the finite low-channel lattice and its residual class modulo  $\mathbb{Z}$ ; they do not prove that the remaining real residual is nonzero along a cofinal family.

**Theorem 5.5** (square-subsequence radius bound). *Let  $t, M, R \in \mathbb{N}$  satisfy*

$$t \geq 2^{32}, \quad M > 0, \quad L_{2t^2} \mid M, \quad M < (R + 1)! - 1.$$

*Then  $3t^3 < 2(R + 1)$ . Consequently no family satisfying these hypotheses for all sufficiently large  $t$  has  $(R(t) + 1)/t^3 \leq 3/2$  eventually, and none has  $R(t) = o(t^3)$ . The historical manuscript cites Erdos68/PaperCompleteExisting.lean, line 236. This module is absent from the retained source pin; this citation does not establish kernel verification at that pin.*

*Proof.* Put  $D = 2t^2$ ,  $k = 2t$  and  $u = D - k + 1 = 2t^2 - 2t + 1$ . Lemma 2.3 at  $N = D$ , with  $n! - 1 \geq n!/2$  and  $\log n! \geq n \log n - n$ , and with  $L_D \leq M < (R + 1)!$ , gives

$$2t(u \log u - u - \log 2) < (R + 1) \log(R + 1) + \binom{2t + 1}{3} \log(2t^2). \quad (21)$$

The historical manuscript cites Erdos68/PaperCompleteSupportedBands.lean, line 25. This module is absent from the retained source pin; this citation does not establish kernel verification at that pin.

Suppose  $R + 1 \leq \frac{3}{2}t^3$ . Since  $t \geq 16$  one has  $\frac{15}{8}t^2 \leq u \leq 2t^2$  and  $\log u \geq 2 \log t$ , so the left side of (21) is at least  $\frac{15}{2}t^3 \log t - 4t^3 - 2t \log 2$ . Using  $\binom{2t+1}{3} < \frac{4}{3}t^3$  and  $\log \frac{3}{2} < \log 2$ , the right side is at most

$$\frac{3}{2}t^3(\log 2 + 3 \log t) + \frac{4}{3}t^3(\log 2 + 2 \log t).$$

Dividing by  $t^3$  and rearranging, (21) would require

$$\frac{1}{3} \log t < 4 + \frac{2 \log 2}{t^2} + \frac{17}{6} \log 2.$$

But  $\log t \geq 32 \log 2$  and  $\frac{2}{3} < \log 2 < 1$  give  $\frac{1}{3} \log t - \frac{17}{6} \log 2 \geq \frac{47}{6} \log 2 > 5$ , while  $4 + 2 \log 2/t^2 < 5$ . This contradiction proves the finite inequality, and the two sequence conclusions follow.  $\square$

The hypotheses concern one specified cancellation architecture. An application to  $S$  must still produce the positive moment and the factorial-size bound from the series, and must show that the channels through  $2t^2$  vanish for the resulting family. The theorem restricts that construction and supplies no irrationality argument.

Theorem 2.4 raises the asymptotic constant in the same estimate.

**Corollary 5.6** (asymptotic radius constant). *Let  $M(t), R(t)$  satisfy  $M(t) > 0$ ,  $L_{2t^2} \mid M(t)$  and  $M(t) < (R(t) + 1)! - 1$  for all sufficiently large  $t$ . Then*

$$\liminf_{t \rightarrow \infty} \frac{R(t) + 1}{t^3} \geq \frac{16}{9}.$$

*Proof.* Put  $r = R(t) + 1$ , so  $L_{2t^2} \leq M(t) < r!$  and  $\log L_{2t^2} < r \log r$ . Theorem 2.4 at  $N = 2t^2$  gives, for every  $\varepsilon > 0$  and all large  $t$ ,

$$\log L_{2t^2} \geq \left(\frac{2\sqrt{2}}{3} - \varepsilon\right) 2\sqrt{2} t^3 (2 \log t + \log 2) \geq \left(\frac{16}{3} - \varepsilon'\right) t^3 \log t.$$

Suppose  $r \leq ct^3$  for arbitrarily large  $t$ , with  $c < 16/9$  fixed. On that subsequence  $r \log r \leq ct^3(3 \log t + \log c) = (3c + o(1))t^3 \log t$ , and  $3c < 16/3$  contradicts the lower bound for large  $t$ .  $\square$

This corollary is asymptotic, and it does not give the strict finite inequality  $R + 1 > \frac{16}{9}t^3$  at every large  $t$ . The finite logarithmic constraint of the channel argument is compatible with equality  $9(R + 1) = 16t^3$  for  $t \geq 4$ , so that constraint alone cannot force a strict finite endpoint at  $16/9$ . The two statements are separate, and Corollary 5.6 rests on the ordinary proof of Theorem 2.4; the finite-block Lean source does not carry the asymptotic deduction, and the asymptotic Lean files are proof candidates rather than kernel verification.

The congruence (18) also explains the prime-channel corrector.

**Theorem 5.7** (two-term prime channel corrector). *Let  $p \geq 3$  be prime and let  $c_{p-1} = p$ ,  $c_p = -1$ , with every other coefficient zero. Then  $M(c) = 0$ ,  $V_p(c) = p! - 1$ , and  $V_d(c) = 0$  for every  $d \geq 2$  with  $d \neq p$ .*

*Proof.* Since  $p \geq 3$ , both indices  $p-1$  and  $p$  lie in the prescribed coefficient domain  $i \geq 2$ . The moment is  $p(p-1)! - p! = 0$ . For  $2 \leq d < p$  the quotient identity  $\lfloor (p-1)/d \rfloor = \lfloor p/d \rfloor$  holds, since  $d \nmid p$ , so the two channel weights carry the same power of  $d!$  and the channel evaluates to  $p(p-1)!/(d!)^{\lfloor p/d \rfloor} - p!/(d!)^{\lfloor p/d \rfloor} = 0$ . For  $d > p$  both indices lie below  $d$ , so both weights are the plain factorials and the same cancellation occurs. At  $d = p$  the weights are  $(p-1)!$  and  $p!/p! = 1$ , giving  $p(p-1)! - 1 = p! - 1$ .  $\square$

For  $p = 2$ , the same auxiliary identities would require  $c_1 = 2$ . They do not define an admissible corrector on the domain  $i \geq 2$ , since every vector supported on that domain has  $c_1 = 0$ .

At zero cost in the moment this supplies a unit in the  $p$ -channel: adding an integer multiple of the corrector to any candidate kernel shifts the  $p$ -channel by multiples of  $p! - 1$  and leaves the moment and every other channel fixed. For every channel rank and every prescribed cutoff there is a factorial-grid kernel together with a remote prime-corrector pair entirely beyond that cutoff, with all requested low channels zero, nonzero moment, and residual in  $[-1/2, 1/2]$ . What is missing is strict nonvanishing: nothing here rules out the rounded residual being exactly zero, and no cofinal family with a strictly nonzero rounded residual has been produced.

## 6 Finite denominator exclusions

Two finite computations constrain a hypothetical denominator  $q$  in  $S = a/q$ , and they constrain different features of it:

$$q \nmid 299999!, \quad q \geq 2^{39990}, \quad q > 10^{12040}. \quad (22)$$

The size bounds hold for every integer  $a$  and positive natural  $q$  with  $S = a/q$ ; no reducedness assumption is needed. They are [kernel-checked in Lean](#). The factorial-grid exclusion  $q \nmid 299999!$  is supported separately by the external GMP computation. It is not a conclusion of the linked Lean theorem; the full factorial-grid certificate remains an outstanding formal obligation.

Neither implies the other. A denominator with no prime factor above 299999 can still fail to divide 299999! by carrying one prime to a high power, and a denominator far below  $2^{39990}$  can still fail that divisibility; a magnitude bound above 299999! would imply it, and no such bound is available here.

The first exclusion comes from an exact interval carry census. An independently implemented GMP integer computation certifies all 299998 carry cells for  $3 \leq m \leq 300000$  with a scale of  $2^{5025679}$  and 96 guard bits, using no floating-point arithmetic; at every step the outward interval is proved to lie inside one half-open unit cell before the carry is recorded. Its unit carries occur exactly at

$$52, 591, 1030, 1407, 1438, 2164, 4258, 10991, 21236,$$

so  $b_{300000} \neq 1$ . Theorem 3.1 then gives  $q \nmid 299999!$  and in particular  $q \geq 300000$ . The receipt is `verification/erdos68-strict-successor.json`, schema `erdos68_strict_successor_exact_interval_v1`, payload digest `2974e8629f5959bdfa2814b63a9522cc9bf46e9e6ab99aaff28a8e980e1a1ee0`, recording the parameters above, the digests of the driver and backend used on its own run, the event-trace digest and the final enclosure. The driver `scripts/check_erdos68_strict_successor.py` and the backend `scripts/check_erdos68_strict_successor_gmp.cpp` accompany it, and their current digests postdate that run, so a byte-identical regeneration of the 300000 receipt is outstanding. The same driver replayed independently at  $m \leq 4000$  reproduces the unit-carry prefix 52, 591, 1030, 1407, 1438, 2164 with a non-unit carry at the endpoint, which gives  $q \geq 4000$  on its own. The census is a separate computation from the Lean development, and the implication it feeds is the kernel-checked one.

The second exclusion has a short integer description. Put  $D = 2^{80000}$  and let  $N$  be the first integer with  $N! - 1 > D$ . Define

$$\ell = \sum_{n=2}^{N-1} \left\lfloor \frac{D}{n! - 1} \right\rfloor, \quad u = \ell + (N - 2) + \left\lfloor \frac{2D}{N! - 1} \right\rfloor + 1.$$

Each rounded prefix term loses less than one, and  $(n + 1)! - 1 > (n + 1)(n! - 1)$  makes the tail from  $N$  smaller than  $2/(N! - 1)$ , so  $\ell/D < S < u/D$ . Here  $N = 7054$  and  $u - \ell = 7053$ . Apply the continued-fraction algorithm to both endpoints at once, retaining a quotient only while the integer parts agree and both remainders are nonzero, and reversing the endpoint order at each inversion. There are 23449 common quotients, and the convergent denominators

$$Q_{-2} = 1, \quad Q_{-1} = 0, \quad Q_j = a_j Q_{j-1} + Q_{j-2}$$

satisfy  $Q_{23448} \geq 2^{39990}$ . Every rational in the open enclosure shares this initial segment, so its reduced denominator is at least  $Q_{23448}$ ; the argument is about a common prefix and does not require the last convergent to lie inside the enclosure. The integer comparison  $2^{39990} > 10^{12038}$  closes (22). The [integer replay](#) and its [receipt](#) record scale 80000 bits, bracket width 7053, 23449 certified partial quotients, power-of-two exponent 39990 and strict decimal exponent 12038.

Both certificates are finite. A refinement of either enlarges the excluded range and returns a statement of the same shape, and neither excludes an eventual unit-carry tail. The continued-fraction method is classical, and no antecedent for these particular numerical outputs was located.

## 7 The remaining arithmetic inputs

The exact target is (8), equivalently (10). The routes below are sufficient conditions for it, and their quantitative hypotheses are unproved. None of them is an equivalent reformulation of the problem.



**Weighted collisions and complementary residues.** On the tailored moving-factor blocks, the local half of (17) is

$$\log \tilde{C}_p < \log \left( \frac{2p^2}{2p+1} \cdot \frac{\prod_{j=p}^{2p-1} j}{q} \right), \quad (23)$$

and the global half is (15). Both must hold on the same cofinal family. For a prime  $r$  absent from the base  $F_p$ , the contribution to  $v_r(\tilde{C}_p)$  counts the levels  $e \geq 1$  at which at least two gaps in  $I_p$  are divisible by  $r^e$ ; for a prime dividing  $F_p$  the base valuation must first be removed, so the layer count is  $v_r(\tilde{C}_p) = \#\{e \geq 1 : h_{r,e+v_r(F_p)}(p) > 1\}$  with  $h_{r,e}(p) = \#\{i \in I_p : r^e \mid i! - 1\}$ . A spacing bound  $h_{r,e}(p)(e+1) \leq 2p + e - 2$  is available. An unweighted count of collisions does not establish (23), and reflected hits may not be discarded, since Wilson reflection produces genuine collision primes. Theorem 12 of Garaev, Luca and Shparlinski [6, arXiv v1, Thm. 12, p. 16] is a multiplicity bound and supplies no lower bound for the complementary residue in (15).

**Repeated-support valuation amplification.** Write  $\Delta_n = u_n/v_n$  in lowest terms, let  $B_n$  be the part of  $n! - 1$  supported on primes occurring in an earlier gap, and put

$$A_n = \prod_{\substack{r \mid B_n \\ v_r(v_n) < v_r(n!-1)}} r^{v_r(n!-1)}.$$

What is established is  $A_n \mid v_{n+1}$  and, when  $A_n > 1$ ,  $u_{n+1} \not\equiv 0 \pmod{A_n}$ . A sufficient quantitative goal, at cofinally many genuinely nonterminal indices  $n$ , is the following with  $m = n + 1$  and  $d = A_n$ :

$$((m+2)m! - 2)v_m \leq m^2(m! - 1)(u_m \bmod d). \quad (24)$$

It implies the upper alternative of (11). A nonzero least representative can equal one while the modulus is enormous, so divisibility and logarithmic mass do not by themselves bound that representative. A lower bound for  $\log A_n$ , and an infinitude of square lifts, are separate possible inputs, and each needs a further implication to reach (24).

**Lower-interval escape.** Prove (11) at arbitrarily large indices; restricting to primes is sufficient. The upper alternative is exactly  $((m+2)m! - 2)v_m \leq m^2(m! - 1)u_m$ . Replacing  $u_m$  by its least nonnegative residue modulo a specified divisor of  $v_m$  gives a stronger sufficient test. Congruence recurrences alone do not count: synthetic models satisfy the available congruences while remaining in the unit-carry branch, and a zero canonical digit is a different event from membership in this Archimedean interval.

**Doubled-prime branch failure.** For an odd prime  $p$ , the carry range and  $Z_{2p} = 2pZ_{2p-1} + 1 - b_{2p}$  give

$$p^2 \mid Z_{2p} \iff \begin{cases} b_{2p} = 1 \text{ and } p \mid Z_{2p-1}, \text{ or} \\ b_{2p} = 1 + p \text{ and } p \mid 2Z_{2p-1} - 1. \end{cases} \quad (25)$$



The historical manuscript cites Erdos68/PaperCompleteSupplementary.lean, line 127. This module is absent from the retained source pin; this citation does not establish kernel verification at that pin.

Reduction modulo  $p$  forces  $b_{2p} \equiv 1 \pmod{p}$ , and the two displayed values are the only possibilities in  $[-1, 2p - 1]$ ; division by  $p$  gives the predecessor conditions. For rational  $S = a/q$  and  $p > q$  the tail estimate gives  $Z_{2p} = (2p)! S$ , which is divisible by  $p^2$ . Ruling out both branches for infinitely many  $p$  therefore proves irrationality. The two conditions must be coupled; controlling only the predecessor residue or only the carry does not meet the hypotheses.

**A remote Cramer residual.** Put  $s_n = ((n+2)!)^2$  and  $i_j = (t+j)s_n$  for  $0 \leq j \leq n+1$ . Form the integer matrix  $A$  with moment row  $(i_j!)_j$  and channel rows  $(W_{d,i_j})_j$  for  $2 \leq d \leq n+2$ , let  $c$  be the cofactor vector of the moment row, and let  $N_d$  be the determinant obtained by replacing that row with  $(W_{d,i_j})_j$ . Cofactor expansion gives  $M(c) = \det A$ ,  $V_d(c) = N_d$ , and  $V_d(c) = 0$  for  $2 \leq d \leq n+2$ . By (18) each  $(V_d(c) - M(c))/(d! - 1)$  is an integer and vanishes for  $d > \max_j i_j$ , so

$$\mathcal{R}_{n,t} := \sum_{d > n+2} \frac{N_d}{d! - 1} = \det(A) S + K_{n,t}, \quad K_{n,t} \in \mathbb{Z}.$$

Seek a family with  $\min_j i_j = t s_n \rightarrow \infty$  and  $\mathcal{R}_{n,t} \notin \mathbb{Z}$ . Such a family proves irrationality: for a fixed denominator  $q$ , every entry of the moment row is divisible by  $q$  once the support is sufficiently remote, so  $q \mid \det A$ . Unbounded pairs  $(n, t)$  do not supply that remoteness, because  $t = 0$  anchors the least support index at zero. The eventual identity  $N_d = \det A$  controls the tail and gives no control over the finite intermediate block, which changes sign; any gcd-of-minors certificate must specify a positive gcd, its rank hypotheses, and its connection to the displayed residual.

**Criteria from the literature that do not apply.** Duverney's fast-series criteria assume two-sided quadratic denominator growth  $cu_n^2 \leq u_{n+1} \leq c'u_n^2$ , whereas  $u_{n+1}/u_n^2 \rightarrow 0$  for  $u_n = n! - 1$ , and his all-positive specialisation additionally requires  $\sum_n |u_{n+1}/u_n^2 - 1| < \infty$ , whose summands tend to one here [9, pp. 275, 285–287]. The theorem of Barreto, Kang, Kim, Kovač and Zhang proves irrationality of  $\sum_n 1/a_n$  under a rapid-growth hypothesis that fails for  $a_n = n! - 1$ , since  $\log(n! - 1) = O(n \log n)$  makes  $(n! - 1)^{1/\psi^n} \rightarrow 1$  for every fixed  $\psi > 1$  [11, Thms. 2–3 and Rem. 4]. Failure of a sufficient criterion is not a theorem about  $S$ , and the useful content is the target these papers identify: a low-height clearing subsequence, or enough exact cancellation in the least common multiple. Theorem 2.4 bounds that cancellation from one side and Theorem 1.2 describes it from the other.

Erdős #68 is open. Every rational representation with positive denominator has  $q \geq 300000$ .

## Sources and evidence

The formal counterparts below are checked by the Lean 4 kernel against Mathlib at source snapshot 92b88dc1bbe0, with no sorry, added axioms, or unchecked evaluation.

Theorem 2.4 and Corollary 5.6 are ordinary proofs given in full above and are not kernel-checked; the associated asymptotic Lean files are proof candidates. The finite-block inequality is the separate Lean source [finite terminal-block inequality](#). The two prefix cancellations, the index-52 example and the continued-fraction enclosure are finite integer calculations with the procedures displayed above. The carry census through 300000 is a separate exact-interval computation with the receipt named in §6.

| Statement                  | Formal counterpart                                                                                                                                                                                                                                               |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Theorem 1.2                | <a href="#">maximal-power survival</a> , with the residue formula at <a href="#">line 129</a> and the denominator form at <a href="#">line 41</a>                                                                                                                |
| Proposition 1.3            | <a href="#">cofinal private hits</a> ; reflection at <a href="#">line 3139</a>                                                                                                                                                                                   |
| Theorem 3.1                | <a href="#">strict-successor misses</a> ; denominator exclusions at <a href="#">line 812</a> and <a href="#">line 836</a>                                                                                                                                        |
| Proposition 3.2            | <a href="#">companion orbit</a>                                                                                                                                                                                                                                  |
| Proposition 3.3            | <a href="#">lower-interval normal form</a> ; finite window at <a href="#">line 122</a> , cofinal consumer at <a href="#">line 145</a> , classification (12) at <a href="#">line 39</a> , radius at <a href="#">line 158</a>                                      |
| Theorem 3.4                | <a href="#">family boundary</a> ; escape form at <a href="#">line 156</a> , member $t = -1$ at <a href="#">line 175</a> , and the irrationality of $e$ at <a href="#">line 211</a>                                                                               |
| Theorem 4.1                | <a href="#">global complementary residue</a> ; coprimality (14) at <a href="#">line 4209</a>                                                                                                                                                                     |
| Equation (17)              | <a href="#">unit-factor scale split</a> , floor identity at <a href="#">line 4701</a> , consumer at <a href="#">line 1574</a>                                                                                                                                    |
| Fixed-owner absorption     | <a href="#">absorbed owner</a>                                                                                                                                                                                                                                   |
| Theorem 5.1 and (18)       | <a href="#">integral channel weight</a> , exact cancellation                                                                                                                                                                                                     |
| Theorem 5.5                | <a href="#">finite radius bound</a> ; sequence form at <a href="#">line 1103</a> , little- $o$ form at <a href="#">line 905</a> , method ceiling ordinary in this record ( <a href="#">sharp_radius_satisfies_square_log_constraint</a> is not in this checkout) |
| Prime channel corrector    | <a href="#">remote reduction</a> , residual identity at <a href="#">line 1559</a>                                                                                                                                                                                |
| Equation (25)              | <a href="#">doubled-prime criterion</a>                                                                                                                                                                                                                          |
| Canonical factorial digits | <a href="#">termination equivalence</a>                                                                                                                                                                                                                          |
| Amplification modulus      | <a href="#">divisibility</a> , nonvanishing at <a href="#">line 5572</a>                                                                                                                                                                                         |

Attribution. Wilson's theorem and the Wilson reflection identity are classical, the latter recorded by Stewart [7, p. 462, (4)]. The factorial-digit termination criterion is Cantor's [4], in the form recorded by Galambos [5, Ch. 1]; the additions here are the telescope  $C = S - e + 2$ , the identification of the digit value  $m - 2$ , and the transport to the shifted family. The multiplicity bound is Garaev, Luca and Shparlinski's [6], and the

lcm deduction from it is not theirs. The survival criterion is a specialisation of Louwsma and Martino's valuation formula [3, Lemma 4.1]. For the Wilson cofinality packaging, the finite radius bound and Theorem 2.4, no antecedent was located, which is a negative search result, and no priority is claimed.

## A Guide to the formal sources

The public `ErdosProblems.Erdos68` package carries the checked source for this note. The declarations below are the dependency chain behind the statements above, together with the subsidiary results developed alongside them; they are pinned to the formal-source commit named at the start of this note.

- `fac floor rat eq cleared`
- `canonical digit eq zero of rational`
- `canonical digit eq floor mul remainder`
- `canonical digit lt radix`
- `canonical remainder recurrence`
- `factorial expansion partial`
- `zero remainder forces zero digit tail`
- `canonical remainder eq zero of zero digit tail`
- `exists rat eq of canonical remainder eq zero`
- `factorial grid plateau`
- `irrational factorial gap series of cofinal nonunit carries`
- `sixty seven le rational denominator`
- `factorial gap zero branch iff lower endpoint cylinder`
- `exists late prefix private factorial gap hit of prime product lt`
- `prime dvd factorial block normalized collision core of odd reflection`
- `one lt factorial block prime hit count of odd reflection`
- `factorial gap large prefix private power modulus dvd tailored block private quotient`
- `factorial gap large prefix private primes unit factor pair tailored block`
- `consecutive unit carries iff positive offset le den`
- `two step den mul transition normalizers`
- `adjacent unit carry window den eq two step den`
- `adjacent unit carry window offset eq two step reduced numerator`
- `factorial moment eq factorial pow mul channel numerator band`

- factorial moment eq zero of channel numerator eq zero band
- factorial moment eq factorial mul channel numerator first band
- exists index ge two mul of factorial moment ne zero of channel eq zero
- square subsequence radius cubic lower
- no eventual square subsequence cubic upper
- channel event eq zero of not dvd
- lambda34 residual bounds
- tendsto factorial gap succ div sq
- not summable abs factorial gap succ div sq sub one
- tendsto factorial gap rpow one div two pow
- pairwise collision core insert gcd lcm
- collision core insert gcd lcm
- collision core div base eq pairwise collision core div gcd
- collision core div base factorization
- collision core div base mul denominator lcm dvd denominator prod
- factorial block fixed pair private quotients eq one
- factorial block normalized collision core le gap prod div gap lcm
- factorial block normalized collision core factorization
- factorial block prime power le gap pow of two hits
- factorial block prime hit pair distance gt exponent
- factorial block prime power hit count mul succ le
- factorial block normalized collision factorization add base lt prime
- factorial block normalized collision factorization add base lt block diameter
- factorial block prime sq gt pred of dvd normalized collision core
- factorial block factor projection lcm eq private modulus
- factorial block complementary factor pair floor le global
- factorial block normalized collision core factorization eq repeated hit layer count
- factorial block normalized collision core factorization eq truncated repeated hit layer count
- factorial block prime power dvd normalized collision core iff one lt hit count of end-point lt base

## B Extended record

The short note above carries the advertised results with complete proofs. This part keeps the surrounding mathematics: the digit kernel the carry and orbit criteria run on, the exact interfaces of the collision-core reduction, the finite certificates, the superseded deductions, and the routes that were tried and closed.

### B.1 The canonical factorial digit kernel

Write  $\theta_0 = \{x\}$  and, for  $m \geq 1$ ,

$$a_m = \lfloor m \theta_{m-1} \rfloor, \quad \theta_m = m \theta_{m-1} - a_m,$$

so that  $\theta_m = \{m!x\}$  for  $m \geq 1$ . The kernel checks the floor formula, the digit bounds  $0 \leq a_m < m$ , the recurrence  $\theta_{m+1} = (m+1)\theta_m - a_{m+1}$ , the finite telescoping expansion

$$x = \lfloor x \rfloor + \sum_{m=2}^N \frac{a_m}{m!} + \frac{\theta_N}{N!},$$

and the rule that a zero remainder at one index forces every later digit to vanish. The rational direction is also checked: if  $q > 0$  and  $q \leq n$ , then  $\text{facFloor}(a/q, n) = ((n!/q) : \mathbb{Z}) a$  and the canonical digit at radix  $n+1$  vanishes, so every rational input has an eventually zero expansion. The converse holds for every real input, and gives

$$x \in \mathbb{Q} \iff a_m(x) = 0 \text{ for all large } m.$$

If all digits after index  $N \geq 1$  vanish, the recurrence gives  $\theta_{N+k} \geq (k+1)\theta_N$ ; since every remainder is below one,  $\theta_N = 0$  and  $x = \lfloor N!x \rfloor / N!$ .

A zero canonical digit is a different event from a zero-branch hit. The returned data contain canonical zero digits at  $m = 5$  and  $m = 23$ , while the zero-branch list is empty through  $m = 100000$ .

A second exact reformulation runs through a defect automaton. For a rational centre recurrence  $F_m = mF_{m-1} + 1 + \varepsilon_m - C_m$ , the kernel checks that the integer ceiling defect code equals  $\lfloor m\delta_{m-1} - \varepsilon_m \rfloor$  and that  $\delta_m = m\delta_{m-1} - \varepsilon_m - q_m$ , with the specialisation  $\varepsilon_m = 1/(m-1)$  written out. What is checked is the algebra of the automaton; proving that the finite-sum residual centre satisfies the premise is a separate step and is not done.

### B.2 Criteria from the literature, in full

Koepef and Schmiersau prove that eventual equality between the floors of  $n$  times a partial sum and  $n$  times its limit forces irrationality [8, Thm. 1.1, p. 117]; their rational-term version obtains that equality from prefix integrality at a scale  $p_n$  together with the strict tail bound  $a - s_n < 1/(np_n)$  [8, Thms. 2.2–2.3, pp. 119–120]. For the natural termwise clearing choice  $p_n = \text{lcm}\{k! - 1 : 2 \leq k \leq n\}$  the last two denominators already obstruct it:

$$p_n \geq \frac{(n! - 1)((n-1)! - 1)}{n-1},$$

because  $\gcd(n! - 1, (n-1)! - 1) = \gcd((n-1)! - 1, n-1) \leq n-1$ . For  $n \geq 4$  the first omitted summand  $1/((n+1)! - 1)$  then already exceeds  $1/(np_n)$ , so this  $p_n$  cannot satisfy their tail

hypothesis. Cancellation in the reduced prefix denominator could give a smaller scale, and proving enough cancellation is another form of the present denominator problem.

Duverney's Theorem 3.1 assumes two-sided quadratic denominator growth  $cu_n^2 \leq u_{n+1} \leq c'u_n^2$ , while  $u_{n+1}/u_n^2 \rightarrow 0$  for  $u_n = n! - 1$  [9, pp. 275, 285–286]; his all-positive Corollary 3.2 additionally requires  $\sum_n |u_{n+1}/u_n^2 - 1| < \infty$ , whereas those summands tend to one [9, Cor. 3.2, p. 287]. Both the ratio limit and the nonsummability are kernel-checked. The ratio proof divides numerator and denominator by  $(n!)^2$ ; the resulting terms tend to zero while  $(1 - 1/n!)^2$  tends to one.

The theorem of Barreto, Kang, Kim, Kovač and Zhang proves irrationality of  $\sum_n 1/a_n$  when  $a_n^{1/2^n} \rightarrow \infty$ , whereas  $(n! - 1)^{1/2^n} \rightarrow 1$  [11, Thms. 2–3, pp. 2–4]; the limit is kernel-checked through the bound  $0 \leq \log(n! - 1) \leq n^2$ . Their proof identifies a useful exact criterion: Mahler's elementary rationality floor is contradicted by prefix-clearing integers  $D_N$  whose cleared positive tails satisfy  $\liminf_N D_N r_N = 0$  [11, Lem. 8 and Prop. 12, pp. 6, 9–12]. The ordinary product of the factorial-gap denominators is far too large for that estimate, so a transfer needs a low-height clearing subsequence, or enough exact cancellation in their least common multiple.

Dividing the strict-successor recurrence  $Z_m = mZ_{m-1} + 1 - b_m$  by  $m!$  and telescoping gives the exact finite identity

$$\frac{Z_M}{M!} = \frac{Z_2}{2!} + \sum_{m=3}^M \frac{1 - b_m}{m!},$$

so the carry defects  $1 - b_m$  are genuine factorial-series coefficients. Hančl and Tijdeman give exact rationality classifications for polynomial coefficients and finite-difference criteria for broader ordinary factorial series [10, Thm. 3.1 and Cor. 3.1, pp. 390–391]; their denominator is the cumulative linear product  $\prod_{n \leq N} (an + b)$ , and the individual number  $N! - 1$  does not occur. Applied to the display above, the Cantor–Oppenheim criterion still needs  $1 - b_m \neq 0$  infinitely often, which is the missing cofinal assertion.

### B.3 A superseded deduction

Before the elementary segment argument of the short note, the record derived a weaker exponent from an external multiplicity theorem. Put  $Q_N = \prod_{2 \leq n \leq N} (n! - 1)$ . For an odd prime  $r$ , let  $m_r$  count the indices  $2 \leq n \leq N$  with  $r \mid n! - 1$ ; such an index satisfies  $n < r$ . The factorial-congruence multiplicity estimate of Garaev, Luca and Shparlinski [6, arXiv v1, Thm. 12, p. 16], applied on  $1 \leq n \leq \min(N, r - 1)$ , gives  $m_r \ll N^{2/3}$ ; the prime 2 divides none of these factors. Writing  $E_r = \max_{2 \leq n \leq N} v_r(n! - 1)$ ,

$$\log Q_N = \sum_r \sum_{n=2}^N v_r(n! - 1) \log r \leq (\max_r m_r) \sum_r E_r \log r \ll N^{2/3} \log L_N,$$

and Stirling summation gives  $\log Q_N \asymp N^2 \log N$ , whence  $\log L_N \gg N^{4/3} \log N$ . Theorem 2.4 supersedes this: its exponent is  $3/2$ , its constant is explicit, and it uses no external theorem. The deduction is retained because it is the shape the returned analyses describe, and because it is the only place a multiplicity bound enters the record.

The primitive lcm divisibility survives cofactor removal: factorial valuations do not remove the channel obstruction once every common cofactor divisor has been removed.

A corank-one cofactor and determinant construction for such primitive kernels has not been formalised, and the divisibility theorem does not establish it.

#### B.4 Plateaux, first exit, and the finite denominator ladder

A second argument works on the rational grid in place of the channels. Let  $H$  be a partial sum and  $q$  a candidate denominator. Writing  $qH = k + r$  and  $q(S - H) = u$ , the next  $q^{-1}$  grid point  $(k + 1)/q$  lies below  $S$  exactly when  $1 \leq r + u$ . The factorial plateau theorem states that if  $H < G \leq S$ , if  $n!G$  is integral, and if  $n!(S - H) < 1$ , then the strict successor of  $n!H$  and the canonical floor of  $n!S$  are the same grid integer. Two rigidity statements follow: consecutive plateau floors, scaled by the next radix, force the canonical digit to vanish; and any first-exit offset  $\delta \in [0, 2)$  with carry  $b = -\lfloor \delta \rfloor$  has  $b \in \{0, -1\}$ , so the exit has exactly two alternatives.

The first-crossing argument continues to a denominator bound. For a rational grid level  $G$  with first crossing  $\tau$  by the literal partial sums, write  $G - H_{\tau-1} = a/v$  with  $a, v > 0$ . Then  $v \geq \tau! - 1$ , and on the  $-1$  exit branch  $v \geq \tau!(\tau! - 1)$ . No coprimality hypothesis on  $a$  and  $v$  is needed.

There is also a direct obstruction at prime indices. With  $\Delta_m$  as in the short note, the kernel checks for  $m \geq 3$  the criterion

$$m \mid Z_m \iff 1 + \frac{1}{m! - 1} < m\Delta_m \leq 2 + \frac{1}{m! - 1},$$

and if  $S = a/q$  with  $q > 0$ , then  $p \mid Z_p$  for every prime  $p > q$ , so one exact missed prime  $p$  gives  $q \geq p$ . Exact kernel reduction gives  $11 \nmid Z_{11}$ , hence  $q \geq 11$ ; exact rational normalisation gives  $60 \nmid Z_{60}$ ,  $64 \nmid Z_{64}$  and  $67 \nmid Z_{67}$ , and the prime index 67 gives the checked bound

$$S = \frac{a}{q}, q > 0 \implies q \geq 67.$$

The exact-interval census of the short note replaces 67 by 300000.

There is a finite peeling identity. For  $x \neq 0, 1$  and  $K \geq 0$ ,

$$\frac{1}{x-1} = \sum_{j=1}^K \frac{1}{x^j} + \frac{1}{x^K(x-1)}.$$

When  $x = k!$  and a chosen factorial scale is divisible by  $(k!)^K$ , the scaled finite sum is integral and only the last term retains the factor  $k! - 1$  in its denominator. The identity isolates one residual fraction before exact bounding, and it supplies no cofinal family of nonzero residuals.

One proposed strengthening is false and is recorded as false: the divisibility  $(m! - 1) \mid \text{den}(V_m)$  fails at the reported strict events  $m = 52$  and  $m = 591$ . Only the Archimedean first-crossing lower bound survives.

There is a second, more arithmetic mechanism at doubled prime indices, stated as (25) in the short note. The formal theorem is not restricted to hand-reduced indices; it is the general unique-slot prime-power criterion specialised to the literal prefixes.

## B.5 The collision-core interfaces

Beyond the definitions used in the short note, the record keeps the exact interfaces of the collision-core reduction.

The collision core has an exact incremental law. For the positive factorial-gap denominators, adjoining  $d_a$  to a finite family  $S$  gives

$$C(S \cup \{a\}) = \text{lcm}(C(S), \text{gcd}(d_a, \text{lcm}_{j \in S} d_j)),$$

since finite-family gcd and lcm distributivity collapses the lcm of all pairwise gcds against  $d_a$  to a single gcd. The same formula holds after adjoining the distinguished base, so each step needs only the old denominator lcm and the old core, with no pairwise rescan.

There is an exact product and lcm bound. If  $\tilde{C}(S)$  is the core after cancelling a positive distinguished base,  $L(S) = \text{lcm}_{j \in S} d_j$  and  $P(S) = \prod_{j \in S} d_j$ , then  $\tilde{C}(S)L(S) \mid P(S)$ , hence  $\tilde{C}(S) \leq P(S)/L(S)$ . For the factorial block this specialises to

$$\tilde{C}_p \leq \frac{\prod_{n \in I_p} (n! - 1)}{\text{lcm}_{n \in I_p} (n! - 1)},$$

an exact bridge from lower estimates for the factorial-gap lcm to upper estimates for the normalised core. It does not close the local scale bound.

The base cancellation is exact prime by prime. With  $F_p = (p-1)!$  and  $C_p$  the unnormalised core,

$$\tilde{C}_p = \frac{\text{lcm}(F_p, C_p)}{F_p} = \frac{C_p}{\text{gcd}(F_p, C_p)}, \quad v_r(\tilde{C}_p) = v_r(C_p) - \min\{v_r(F_p), v_r(C_p)\}.$$

So  $r^e \mid \tilde{C}_p$  exactly when the pairwise core carries  $r^{e+v_r(F_p)}$ , which in the block forces two distinct gaps to be divisible by that higher power. The sharp surviving valuation cap  $v_r(\tilde{C}_p) + v_r(F_p) < r$  then gives  $p-1 < r(r-1) < r^2$  for every support prime, and  $\tilde{C}_p$  is coprime to  $k!$  whenever  $k(k-1) \leq p-1$ . This removes every factorial channel below the moving square-root cutoff, and it does not bound the aggregate product of the remaining large prime powers.

For collision estimates that already provide an upper-half hit, no exponent is lost to normalisation: if  $r$  divides a displayed gap at some  $n \geq p$ , then  $r \nmid F_p$  and  $r^e \mid \tilde{C}_p \iff r^e \mid C_p$  for every  $e > 0$ . This has an exact incidence-count form,

$$r^e \mid \tilde{C}_p \iff 1 < \#\{i \in I_p : r^e \mid i! - 1\},$$

so a source estimate giving at most one  $r^e$ -hit deletes that exponent and yields  $v_r(\tilde{C}_p) < e$ . The local aggregation is exact:

$$v_r(\tilde{C}_p) = \#\{e \in [1, r-1] : 1 < \#\{i \in I_p : r^e \mid i! - 1\}\},$$

and for an endpoint prime with  $2p-1 < r$  the range truncates to  $e \in [1, 2p-4]$ . The same equivalence holds under the exact condition  $r \nmid F_p$  in place of the endpoint inequality,



which covers the entire moving prime range at and above the block parameter; at  $e = 2$  it gives a conditional squarefree conclusion  $v_r(\tilde{C}_p) \leq 1$ .

The local load has a distance-sensitive witness. If  $r$  is prime,  $e > 0$  and  $r^e \mid \tilde{C}_p$ , there are  $i < j$  in  $I_p$  with  $r^{e+v_r(F_p)} \mid i! - 1$ ,  $r^{e+v_r(F_p)} \mid j! - 1$  and  $r^{e+v_r(F_p)} \leq j^{j-i}$ ; consequently  $(2p-1)^d < r^{e+v_r(F_p)}$  forces  $d < j-i$ . The spacing hypothesis is discharged internally: any two  $r^e$ -hits  $i < j$  satisfy  $e < j-i$ , because  $r \mid j! - 1$  already forces  $j < r$  and the gap-power inequality converts that size relation into strict separation. The pairwise form is stronger than the selected-witness form: for arbitrary  $r, e$  and any displayed hits  $i < j$ ,

$$r^e \mid i! - 1, \quad r^e \mid j! - 1 \quad \implies \quad r^e \leq j^{j-i},$$

so every prime-power hit layer is an  $e$ -separated subset of the block and

$$(e+1) \#\{i \in I_p : r^e \mid i! - 1\} \leq 2p + e - 2.$$

The unweighted packing step is therefore complete. Globally,

$$r \mid \tilde{C}_p \quad \implies \quad v_r(\tilde{C}_p) + v_r(F_p) < 2p - 3$$

for every prime  $r$  and  $p \geq 2$ , so even primes already present in the normalisation base pay for their base valuation inside the same block-diameter budget. What remains is to bound the layer counts uniformly as  $r$  and  $p$  move, multiply the surviving contributions, and still close the complementary-residue coordinate.

The squarefreeness premise is not proved. An exhaustive modular scan through  $r \leq 2,000,000$  and  $n \leq 240$  found four individual square hits and no prime with two such hits. All 498,501 pairs  $2 \leq a < b \leq 1000$  have squarefree  $\gcd(a! - 1, b! - 1)$ , and the aggregate squarefree-collision scan through  $p = 499$  stays below 0.374 of the upper-descending-factorial logarithmic scale. These are finite exact computations. They carry neither theorem authority nor an asymptotic incidence bound.

A finite variant of the cofinal private supply compares the product of a chosen set of primes, each at least 5, with  $\prod_{2 \leq k \leq B} (k! - 1)$ ; if the prime product is larger, at least one chosen prime has no hit through  $B$ , while Wilson still bounds its least hit by  $q - 2$ . Wilson reflection limits what a linear-size divisor can be assumed to be: if  $n$  is odd,  $n < q$ , and  $q \mid n! - 1$ , then  $q \mid (q - n - 1)! - 1$ , and when both indices lie in one block with the reflected hit earlier, equivalently  $q < 2n + 1$ , the repeated hit survives predecessor-factorial normalisation and its full-block incidence count exceeds one.

Stewart states that for every  $\varepsilon > 0$  there are infinitely many odd  $n$  whose least prime factor  $q$  of  $n! - 1$  satisfies

$$n < q < \left( \frac{\sqrt{145} - 1}{8} + \varepsilon \right) n,$$

the printed text transferring estimate (9) from  $n! + 1$  to  $n! - 1$  [7, p. 464]. The source controls  $q$  relative to the later index  $n$  and supplies no control relative to the private first-hit index, so it is collision-core input. The missing private-anchor and global product estimates are elsewhere.

One selected prime  $q$  already furnishes the coprime factor pair  $(1, q)$ : its projection moduli are  $R_p$  and  $R_p/q$ , whose least common multiple is  $R_p$ , so no second selected

prime is needed and there is no hidden equality-or-disagreement branch in that specialisation. The underlying projection rigidity is general: if the endpoint numerator  $Z$  is congruent to a weighted numerator  $T$  modulo  $R$ , then every divisor  $Q$  of  $R$  with  $Z \leq B < Q$  satisfies  $T \bmod Q = Z$ , so two divisors above  $B$  with unequal projected residues exclude that bounded endpoint, and unequal projections force  $\min(Q_1, Q_2) \leq T$ .

## B.6 Finite channel and carry certificates

The finite-support vector  $\lambda = 2e_3 - e_4$  has, by kernel check,  $V_2(\lambda) = 0$ , factorial moment  $-12$ ,  $V_3(\lambda) = -2$ ,  $V_4(\lambda) = 11$ , and  $V_d(\lambda) = -12$  for every  $d \geq 5$ . Under the exact rational tail enclosure  $1/119 < \Theta_4 < 1/50$  its residual lies strictly between  $-93/575$  and  $-309/13685$ , so it is nonzero and subunit.

Exact integer regeneration verifies the canonical primitive kernels for every  $2 \leq D \leq 12$ : channels 2 through  $D$  vanish, the factorial moment is  $L_D$ , and the coefficient content is one. At  $D = 9$  the moment is  $L_9 = 31540008254514077395$  and, after the stated prime-unit shift,  $1353/100000 < R_9 < 1354/100000$ . At  $D = 3$  the vector  $c = (-40, 55, -10, 1)$  on the support  $(3, 4, 5, 6)$  annihilates channels 2 and 3, has moment 600, and satisfies

$$0.09925341997208298 < L_3(c) < 0.09925341997208300,$$

which excludes denominators dividing 600.

There are two different computations at the same endpoint. A returned interval computation reports the stronger geometric statement that no zero-branch event occurs at any  $m \leq 100000$ ; its cited executable and source digest were not supplied, so that classification remains external finite evidence. The strict-successor carry computation used in the short note is local and independently regenerated, with its own source, backend, payload and receipt digests.

## B.7 Limits of fixed-coordinate arguments

- Residue vectors, their recurrences, and window widths admit synthetic all-hit blocks, so they cannot prove irrationality on their own.
- Known pointwise prime congruences, prime-dilation congruences, parity, and the exact prime coefficient formula admit a synthetic rational countermodel.
- Wilson quotients, harmonic sums,  $p$ -adic gamma identities, and factorial residues do not control the required Archimedean floor without an additional coupling theorem. Every prime-window test factors into a sharp Archimedean strict-ceiling condition and a modular divisibility condition, and the missing ingredient is the coupling between them.
- Fixed-denominator scalar canonical-product localisers and rank-saturated consecutive-jet Hermite–Padé systems pay the full factorial-gap denominator. For  $E(z) = \prod_{n \geq 2} (1 - z/n!)$  the genus-zero product satisfies  $-E'(1)/E(1) = S$ , and the natural scalar linear form carries the coefficient  $Q_N = \prod_{2 \leq n \leq N} (n! - 1)$ , for which  $Q_N$  times the tail diverges. Exact first-order interpolation, scalar residue weighting, the natural Wronskian, and rank-saturated consecutive jets all reassemble the same prohibitive denominator.

- Zero-moment variations cannot create an additional fractional cancellation coordinate, and factorial valuations cannot absorb the channel lcm obstruction.
- A fixed pair of low-index private owners cannot make the projection argument cofinal, by the absorption statement of the short note.

### B.8 The generic shape behind the lower-interval form

The cofinal statement in the short note has a generic ancestor. For a real  $x$  and any positive sequence  $c_m$  with  $c_m < 1/m$  eventually,

$$x \notin \mathbb{Q} \iff \{(m-1)!x\} \geq c_m \text{ for arbitrarily large } m.$$

For rational  $x$  the fractional parts vanish eventually; if all large fractional parts lie below  $c_m$ , multiplying by  $m$  stays below one, so the next canonical digit is zero, and eventual zero digits force rationality. The specific target  $c_m = E_m/m$  is useful because it translates into finite predecessor-gap conditions. The smallness of the target alone establishes no metric, measure or equidistribution theorem for this orbit, and none is claimed.

## C Complete result-family map

This section places every registered family for this problem in the shared 70-family reader order. The five display bands control exposition only; the separate promotion state currently covers 6 families and is reported but does not hide or strengthen any family. Mathematical statements and evidence modes come from the public claim registry. Across all eight problems the public result-atom catalog contains 681 exact packet coordinates; this problem contributes 112. The recovered source catalog contains 682 rows; 1 row(s) belong to families absent from the current claim registry and remain disclosed as detached source rows. Catalog rows expose bounded statement excerpts plus full-source digests, not a claim that every complete packet statement is reproduced here.

### C.1 Factorial carry characterisation

**Reader position.** 1 of 70; display band: front door. Formal editorial disposition: promote. These are separate classifications.

**Reader entry.** Exact equivalence between irrationality, cofinally many non-unit carries, and cofinal strict-successor divisibility failures.

Exact equivalence between irrationality, cofinally many non-unit carries, and cofinal strict-successor divisibility failures.

**Authority and reach.** Lean kernel; Comparator-selected; locally proved result; novelty unassessed.

**Exact boundary.** An exact reformulation does not supply the required cofinal failures.

**Result-atom population.** 33 of 681 public coordinates. The public atom catalog groups them under this family.

**Comparator assurance.** exact selected interface; 2 executable interface(s), 2 repository-registered selected result interface(s).

**Publication placement.** This family remains in the complete long record and is not a short-note headline.

- `ErdosProblems.Erdos68.irrational_factorialGapSeries_iff_cofinal_nonunit_carries`
- `ErdosProblems.Erdos68.irrational_factorialGapSeries_iff_cofinal_strictFacTopRat_misses`

### C.2 Factorial channel and projection rigidity

**Reader position.** 23 of 70; display band: mechanism. Formal editorial disposition: split. These are separate classifications.

**Reader entry.** Exact low-channel classification, the attainable moment ideal with a content-one primitive generator, full-residual transparency modulo integers, finite quotient-band factorisation and breakpoint, channel congruences, a two-term prime-channel corrector, and endpoint-weighted projection rigidity.

Exact low-channel classification, the attainable moment ideal with a content-one primitive generator, full-residual transparency modulo integers, finite quotient-band

factorisation and breakpoint, channel congruences, a two-term prime-channel corrector, and endpoint-weighted projection rigidity.

**Authority and reach.** Lean kernel; locally proved result; novelty unassessed.

**Exact boundary.** These structural results determine the finite low-channel moment lattice and the residual class modulo integers, but they do not produce a cofinal nonintegral real residual or another obstruction settling Erdős 68.

**Result-atom population.** 63 of 681 public coordinates. The public atom catalog groups them under this family.

**Comparator assurance.** represented by selected interface; 0 executable interface(s), 0 repository-registered selected result interface(s).

**Publication placement.** This family is also admitted to the short note.

- Erdos68.factorialMoment\_eq\_factorial\_pow\_mul\_channelNumerator\_band
- Erdos68.exists\_index\_ge\_two\_mul\_of\_factorialMoment\_ne\_zero\_of\_channel\_eq\_zero
- ErdosProblems.Erdos68.PaperComplete.attainable\_moment\_ideal
- ErdosProblems.Erdos68.PaperComplete.exact\_moment\_ideal\_with\_primitive\_attainment
- ErdosProblems.Erdos68.PaperComplete.minimum\_moment\_content\_one
- ErdosProblems.Erdos68.PaperComplete.minimumMoment\_independent\_prime
- ErdosProblems.Erdos68.PaperComplete.residual\_transparency
- ErdosProblems.Erdos68.PaperComplete.summable\_fullResidual
- ErdosProblems.Erdos68.PaperComplete.zero\_moment\_residual\_integral
- ErdosProblems.Erdos68.PaperComplete.equal\_moment\_residual\_integer\_difference
- Erdos68.channelNumerator\_mod\_factorialMoment
- Erdos68.primeTranslator\_channelResidual\_eq\_one
- ErdosProblems.Erdos68.projection\_disagreement\_excludes\_bounded\_endpoint

### C.3 Factorial conditional producers

**Reader position.** 43 of 70; display band: frontier. Formal editorial disposition: hold. These are separate classifications.

**Reader entry.** Cofinal zero-branch or large private-factor hypotheses imply the required irrationality conclusion.

Cofinal zero-branch or large private-factor hypotheses imply the required irrationality conclusion.

**Authority and reach.** Lean kernel; conditional reduction.

**Exact boundary.** The producer hypotheses are unproved.

**Result-atom population.** 1 of 681 public coordinates. The public atom catalog groups them under this family.

**Comparator assurance.** not selected for comparator; 0 executable interface(s), 0 repository-registered selected result interface(s).

**Publication placement.** This family is also admitted to the short note.

## C.4 Factorial lcm growth

**Reader position.** 48 of 70; display band: frontier. Formal editorial disposition: hold. These are separate classifications.

**Reader entry.** An elementary terminal-block argument gives  $\liminf \log L_N / (N^{3/2} \log N) \geq 2\sqrt{2}/3$  for the lcm of factorial-gap denominators. The displayed bound needs no external multiplicity theorem.

An elementary terminal-block argument gives a lower bound of order  $N^{3/2} \log N$  for the lcm of factorial-gap denominators. The displayed bound is ordinary mathematics. An older, weaker multiplicity-based route is historical context only. The linked asymptotic Lean files are proof candidates and are not cited as kernel verification. The finite-block inequality has a separate Lean source.

**Authority and reach.** ordinary mathematical argument.

**Exact boundary.** The LCM asymptotic is an ordinary paper proof, not a Lean-checked theorem. Comparator cannot certify the authored deduction. The bound does not decide irrationality.

**Result-atom population.** 3 of 681 public coordinates. The public atom catalog groups them under this family.

**Comparator assurance.** not applicable to comparator; 0 executable interface(s), 0 repository-registered selected result interface(s).

**Publication placement.** This family is also admitted to the short note.

## C.5 Factorial finite certificates

**Reader position.** 59 of 70; display band: technical support. Formal editorial disposition: hold. These are separate classifications.

**Reader entry.** Lean checks small exact misses; a hash-bound GMP scan reaches index 300000 and yields the corresponding finite denominator exclusion.

Lean checks small exact misses; a hash-bound GMP scan reaches index 300000 and yields the corresponding finite denominator exclusion.

**Authority and reach.** Lean finite theorem plus external exact certificate; finite computation.

**Exact boundary.** Finite computation does not change the cofinal quantifier.

**Result-atom population.** 12 of 681 public coordinates. The public atom catalog groups them under this family.

**Comparator assurance.** not applicable to comparator; 0 executable interface(s), 0 repository-registered selected result interface(s).

**Publication placement.** This family is also admitted to the short note.

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